



ASSIGNMENT 2

PROBABILITY & STATISTICAL DATA ANALYSIS

(SECI1143-03)

NAME	MATRIC NUMBER
MUHAMMAD AFIQ DANIAL BIN ROZAIDIE	A23CS0117
MOHAMED ALIF FATHI BIN ABDUL LATIF	A23CS0112
MUHAMMAD NAIM BIN ABDULLAH	A23CS0134
AHMAD ADIB ZIKRI BIN A.MAZLAM	A23CS0205

LECTURER: DR NOR HAIZAN BINTI MOHAMED RADZI

Question 1

class interval	midpoint	Frequency	Cumulative frequency	fX_i
$150 \leq x < 160$	155	12	12	$155 \times 12 = 1860$
$160 \leq x < 170$	165	20	32	3300
$170 \leq x < 180$	175	5	37	875
$180 \leq x < 190$	185	3	40	555
Total		40		6590

a) Mean

$$\bar{X} = \frac{\sum_{i=1}^k f_i X_i}{n} = \frac{6590}{40} = 164.75$$

b) Median

$$\frac{N}{2} = \frac{40}{2} = 20$$

$\therefore$ median class = $160 \leq x < 170$

$$L = 160$$

$$W = 10$$

$$N = 40$$

$$f_{med} = 20$$

$$cf_p = 17$$

$$\text{median} = L + \frac{\frac{N}{2} - cf_p}{f_{med}} (W) = 160 + \frac{20 - 17}{20} (10) = 161.5$$

c) Mode

$$l = 160$$

$$\text{mode} = l + h \times \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)}$$

$$h = 10$$

$$(2f_1 - f_0 - f_2)$$

$$f_1 = 20$$

$$= 160 + 10 \times \frac{(20 - 12)}{(40 - 12 - 5)}$$

$$f_0 = 12$$

$$(40 - 12 - 5)$$

$$f_2 = 5$$

$$= 163.418$$

d) Modal class

$$\therefore 160 \leq x < 170$$